

Deliverables & Academic Verification Requirements

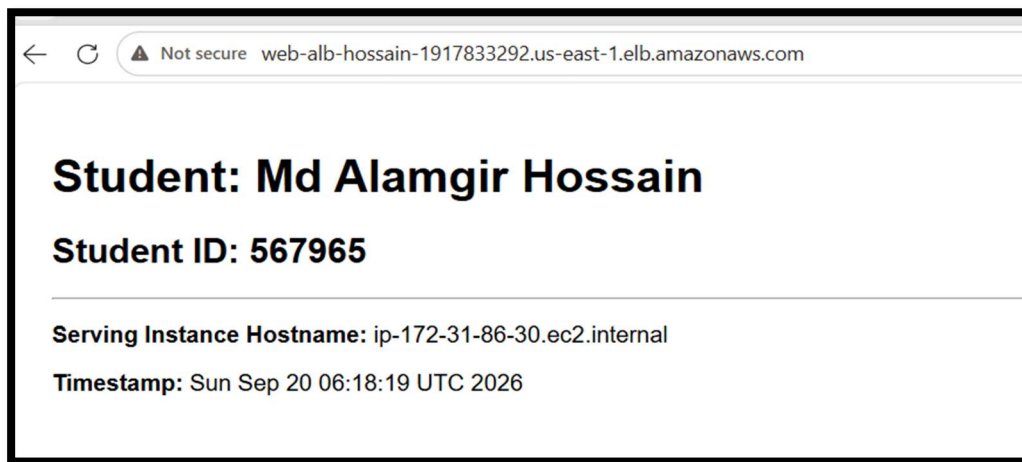
To verify authentic lab completion, all submitted screenshots **must capture the entire uncropped desktop or browser window**, showing:

1. The **top-right corner of the AWS Console** displaying your unique **Account ID** and **IAM Username/Federated Role**.
2. The browser address bar with the complete ALB DNS address visible.

Required Screenshots

Screenshot 1 (ALB Routing - Instance A): The browser rendered via the ALB showing your Name, Student ID, and Hostname A.

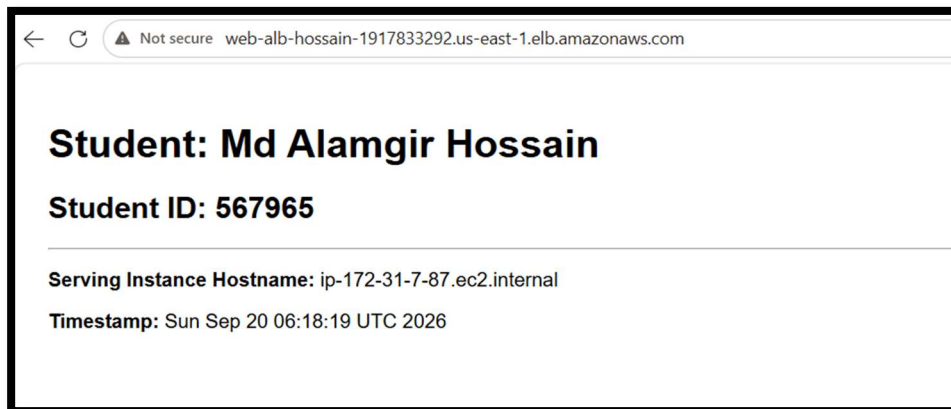
Required Screenshot Instance A



Note: Provides the required ALB response with the student details and the first recorded hostname.

Screenshot 2 (ALB Routing - Instance B): The browser refreshed via the ALB showing the exact same Name and Student ID, but with Hostname B.

Required Screenshot Instance B



Note: Provides the required ALB response with identical student details and a different hostname.

Screenshot 3 (EC2 ASG Self-Healing): The EC2 Instances dashboard showing one instance Terminated and the new replacement instance in a Running state, with the AWS Account ID visible in the console header.

ASG Self-Healing Evidence

The screenshot shows the AWS Management Console for the EC2 Instances dashboard. The top navigation bar includes the AWS logo, a search bar, and the account ID 'hossain8078 (841452566550)'. The left sidebar shows the 'EC2' menu with options like 'Dashboard', 'AWS Global View', 'Events', 'Instances', 'Instance Types', 'Launch Templates', 'Spot Requests', 'Savings Plans', 'Reserved Instances', 'Dedicated Hosts', 'Capacity Reservations', 'Capacity Manager', and 'Images'. The main content area displays the 'Instances (3)' table. The table has columns: Name, Instance ID, Instance state, Instance type, Status check, and Availability. The instances listed are:

Name	Instance ID	Instance state	Instance type	Status check	Availability
	i-0350d70f5eee1d0b4	Running	t2.micro	2/2 checks passed	us-east-1b
	i-0f545c1e4c60c3aa9	Terminated	t2.micro	-	us-east-1b
	i-01961bb4ec97ca8c4	Running	t2.micro	2/2 checks passed	us-east-1a

Below the table, the 'All instances' section shows three cards: 'Instance alarm states' (In alarm: None), 'Service health' (EC2 service status: Operating normally), and 'Health events affecting EC2 instances' (Open events: None, Upcoming events: None).

Note: Shows the terminated original instance together with its replacement and the surviving instance.

Final EC2 Recovery State

The screenshot shows the AWS Management Console for the EC2 Instances dashboard. The top navigation bar includes the AWS logo, a search bar, and the account ID 'hossain8078 (841452566550)'. The left sidebar shows the 'EC2' menu with options like 'Dashboard', 'AWS Global View', 'Events', 'Instances', 'Instance Types', 'Launch Templates', 'Spot Requests', 'Savings Plans', 'Reserved Instances', 'Dedicated Hosts', 'Capacity Reservations', 'Capacity Manager', and 'Images'. The main content area displays the 'Instances (3)' table. The table has columns: Name, Instance ID, Instance state, Instance type, Status check, Availability Zone, and Public IPv4 DNS. The instances listed are:

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
	i-0350d70f5eee1d0b4	Running	t2.micro	2/2 checks passed	us-east-1b	ec2-44-204-158-84.co...
	i-0f545c1e4c60c3aa9	Terminated	t2.micro	-	us-east-1b	-
	i-01961bb4ec97ca8c4	Running	t2.micro	2/2 checks passed	us-east-1a	ec2-44-211-187-236.co...

Note: Confirms Auto Scaling restored the desired capacity with two running instances.

Conceptual Analysis Questions

1. Troubleshooting Network Protocols

When the ALB DNS name is entered without `http://`, a modern browser may automatically attempt an HTTPS connection on port 443. However, this lab configures the Application Load Balancer with only an HTTP listener on port 80. The security group also permits HTTP traffic on port 80 but does not permit HTTPS traffic on port 443.

Therefore, the load balancer is not listening for HTTPS connections, and the security group does not allow them. The HTTPS request cannot reach the web application, so the browser may eventually display `ERR_TIMED_OUT` or another connection error. Entering `http://` explicitly tells the browser to connect through port 80, which matches both the ALB listener and the security-group rule.

2. Configuration Management

Using EC2 user data is architecturally superior to configuring each server manually through SSH because it automates and standardizes the server-configuration process. Every instance created from the launch template automatically installs Apache, starts the web service, enables it at startup, and creates the required webpage.

This approach provides several advantages:

- All instances receive the same configuration.
- New and replacement instances can become operational automatically.
- It reduces human error and configuration differences between servers.
- It supports rapid scaling because administrators do not need to configure every new instance manually.
- The configuration is repeatable, documented, and easier to update.
- Administrators do not need to maintain direct SSH access for routine deployment.

In this lab, the user-data script allowed the ASG replacement instance to become a working web server without manual intervention.

3. High Availability and Health Checks

The target group performs periodic HTTP health-check requests to the configured health-check path, which is `/` in this lab, using port 80. If an instance responds successfully with the expected HTTP status code, normally `200`, it is considered healthy. If the target repeatedly fails to respond or returns an unacceptable response, the target group marks it unhealthy.

The ALB stops routing normal user requests to an unhealthy target and sends traffic only to healthy instances. Because Elastic Load Balancing health checks were enabled for the Auto Scaling Group, the ASG can also recognize the unhealthy instance. It then terminates the unhealthy instance, launches a replacement from the launch template, attaches the replacement to the target group, and waits for it to pass the health checks. This process restores the ASG to its desired capacity of two instances.

4. Real-World Business Application

An online retail company would benefit from automatically scaling its ASG based on CloudWatch metrics. During normal periods, a small number of servers may be sufficient. However, during Black Friday, holiday promotions, or flash sales, website traffic can increase quickly.

A target-tracking policy could monitor metrics such as average CPU utilization or Application Load Balancer request count per target. When demand rises above the configured target, the ASG launches additional instances, up to its maximum capacity, to maintain performance and prevent the website from becoming overloaded. When traffic decreases, the ASG removes unnecessary instances.

Automatic scaling provides sufficient capacity during periods of high demand while reducing infrastructure costs during quiet periods. A static desired capacity of two might be insufficient during traffic spikes and unnecessarily expensive when demand is low.